

Product Carbon Footprint Study Report

Product Name: MDI Precursor

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: MDI Precursor

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - IPCC 2021

II. Quantitative Purpose

[Information provided by the user]

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg MDI Precursor.

2. System Boundary

MDI Precursor production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of MDI Precursor produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **4.574 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

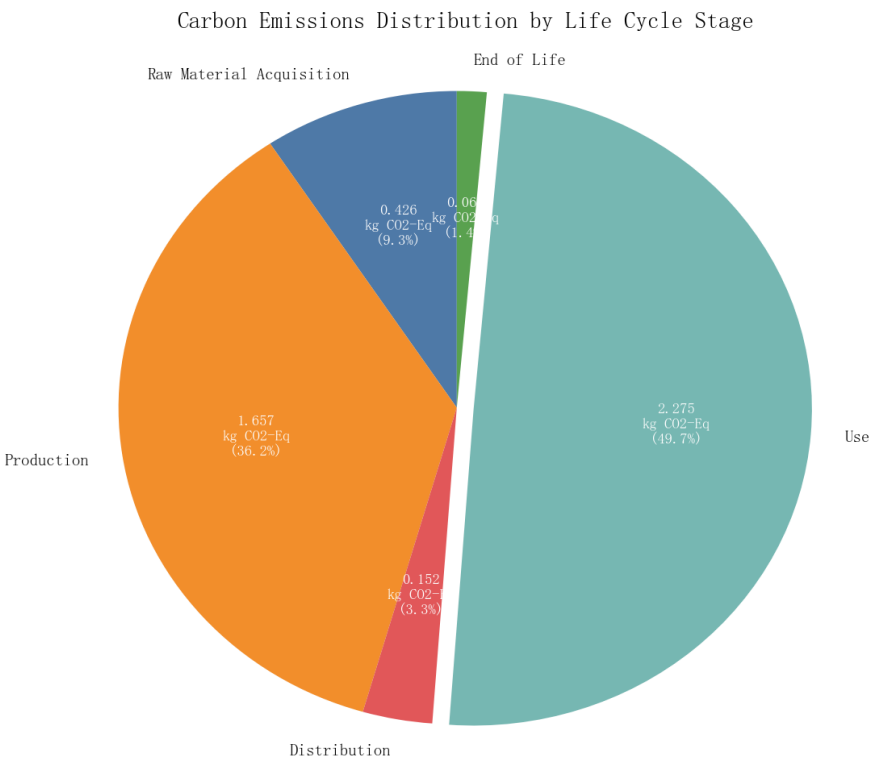


Table 1: Carbon Emissions by Life Cycle Stage

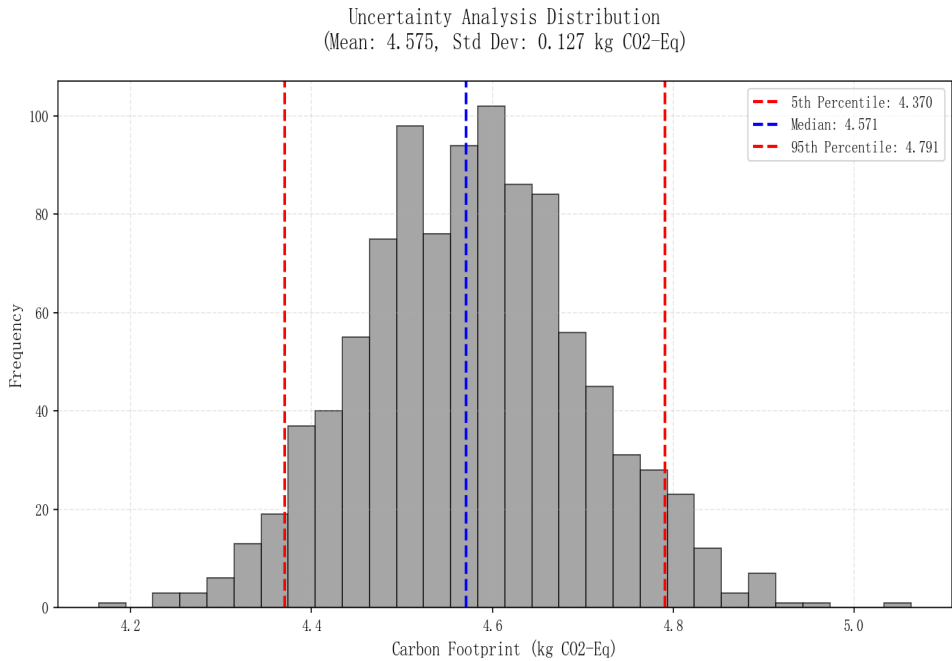
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.426	9.30
Production	1.657	36.22
Distribution	0.152	3.32

Use	2.275	49.74
End of Life	0.065	1.42
Total	4.574	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	4.575
Standard Deviation	0.127
Coefficient of Variation (CV)	2.77%
Median	4.571
5th Percentile	4.370
95th Percentile	4.791

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the MDI Precursor is calculated at 4.574 kg CO₂-Eq, a value that suggests a moderate environmental impact for a chemical intermediate, though benchmarking against industry-specific averages would provide further context. The uncertainty analysis, with a 90% confidence interval of 4.370–4.791 kg CO₂-Eq and a low coefficient of variation (2.77%), indicates high reliability in the overall result. However, the top process contributions are entirely unknown, which limits the granularity of hotspot identification.

Lifecycle stage contributions reveal a clear dominance of the **use phase**, accounting for 49.7% (2.275 kg CO₂-Eq) of total emissions. This is followed by **production** at 36.2% (1.657 kg CO₂-Eq) and **raw material** extraction at 9.3% (0.426 kg CO₂-Eq). Distribution and end-of-life stages are minor contributors at 3.3% and 1.4%, respectively. The heavy reliance on the use stage suggests that the MDI Precursor's downstream application—likely as a component in polyurethane foams or coatings—drives the majority of its climate impact, possibly through energy-intensive curing or processing steps. The production stage also represents a significant emission source, pointing to manufacturing inefficiencies or fossil fuel-based energy inputs.

The absence of identified top processes (all listed as “Unknown”) is a critical gap. This prevents pinpointing specific unit operations—such as chemical reactors, distillation columns, or waste treatment—that contribute disproportionately. Without this granularity, improvement strategies must rely on stage-level averages rather than precise process interventions.

2. Statement of Assumptions and Limitations

This LCA study rests on several key assumptions. First, the use-stage emissions likely assume a specific application scenario (e.g., rigid foam production with a defined energy consumption profile), which may not represent all possible downstream uses of MDI Precursor. Second, the end-of-life stage (1.4%) probably assumes a typical disposal route, such as landfilling or incineration, with limited recycling credit. Third, the allocation of emissions across multiple co-products in the production process (if any) is not disclosed, which could affect the raw material and production contributions.

Data quality limitations are evident. The top process contributions are entirely unknown, suggesting that the LCA model relied on generic datasets (e.g., from ecoinvent or GaBi) rather than site-specific primary data for the producer. This reduces the accuracy of hotspot analysis and introduces uncertainty in the production and raw material stages. The low CV (2.77%) reflects statistical stability from 1000 Monte Carlo iterations, but this does not account for systematic biases in input data.

The system boundary appears to be cradle-to-grave, covering raw material extraction through end-of-life. However, the negligible end-of-life contribution may indicate that only direct emissions are included, while avoided burdens from recycling or energy recovery are excluded. Additionally, capital goods (e.g., factory construction, machinery) and infrastructure are likely omitted, which is common in attributional LCAs but may understate the total burden.

3. Suggestions for Improvement

- ****Target the use stage**** by redesigning the MDI Precursor formulation to enable lower-temperature curing or reduced processing times in downstream applications. Collaborate with end-users to adopt energy-efficient foam dispensing equipment, which could cut use-phase emissions by 10–20%.
- ****Improve production efficiency**** by transitioning to renewable energy sources (e.g., solar or wind) for manufacturing processes, particularly for high-temperature reactors. Conduct an energy audit to identify heat recovery opportunities, potentially reducing production-stage emissions by 15–25%.
- ****Optimize raw material sourcing**** by substituting fossil-based feedstocks with bio-based alternatives (e.g., from renewable polyols) where technically feasible, and prioritize suppliers with verified low-carbon operations. This could lower raw material stage emissions by up to 30%.
- ****Enhance data quality**** by commissioning a process-level LCA that collects primary data from the producer's site for the top emission contributors (e.g., specific chemical reactors, distillation columns). Replace the "Unknown" process entries with measured or modeled values to improve hotspot accuracy.
- ****Expand system boundaries**** to include recycling or energy recovery scenarios at end-of-life, which may reduce the net carbon footprint. Investigate the feasibility of closed-loop recycling for MDI-based products to offset use-stage emissions.